

TechnoSys: A Web-Based HRIS with Mobile Geofenced Biometrics Model for Field Attendance Management

BACKGROUND OF THE STUDY

- **Company Profile:** Technocycle Corporation – HVAC/R service contractor (installation, repair, maintenance, duct fabrication).
- **Target Population:** Focuses on the Main Branch in Pacita, San Pedro, Laguna (33 active employees)
- **Operational Transition:** Moving Technocycle Corporation from vulnerable standalone biometrics, paper forms, and Messenger group chats to an automated, tamper-proof HRIS ecosystem.

STATEMENT OF THE PROBLEM

Main Problem:

How can TechnoSys be developed and evaluated to automate field attendance, payroll, equipment tracking, and dispatching for Technocycle Corporation?

Research Questions:

1. **Existing Challenges:** Baseline operational challenges in attendance, ₱200 late penalties, payroll delays, tool custody, leaves, and Messenger scheduling.
2. **System Features:** Engineering Geofenced Biometrics, Statutory Payroll, Tool Tracker, Digital Leaves, and Dispatch Scheduler.

STATEMENT OF THE PROBLEM

3. **System Effectiveness:** Measuring record accuracy, fraud prevention, tool traceability, and field mobile usability.
4. **Software Quality:** Evaluation using the ISO/IEC 25010 Standard (Suitability, Efficiency, Usability, Reliability, Security, Maintainability).
5. **System Enhancements:** Formulating continuous operational recommendations.

SCOPE

- Develop a web-based HRIS for Technocycle Corporation (Main Branch – Pacita, San Pedro, Laguna).
- The study involves 33 employees

The system includes:

- GPS Geofencing & Biometric Attendance
- Employee Records
- Mobile Attendance Application
- Real-Time Attendance Monitoring
- Automated Payroll Computation
- Attendance Reports & Dashboard
- Service Ticket Assignment & Field Deployment Tracking
- Role-Based Access Control

DELIMITATION

- Covers only the main branch of Technocycle Corporation.
- Does not include other branches unless expanded in the future.

Geofencing accuracy depends on:

- GPS Signals
- Internet Connection
- Mobile Device Capability
- Biometric issues caused by device hardware (e.g., fingerprint sensor or camera problems) are outside the scope of the system.

SIGNIFICANCE OF THE STUDY

- **Technocycle Corporation Management**
 - Supports better workforce monitoring, operational efficiency, and informed decision-making.
- **HR and Payroll Personnel**
 - Automates attendance and payroll processes while reducing manual work and human errors.
- **Field Technicians and Employees**
 - Provides secure attendance validation, transparent payroll records, and minimizes attendance disputes.

SIGNIFICANCE OF THE STUDY

- **Future Researchers**
 - Provides a foundation for future studies related to HRIS, attendance automation, and digital transformation.
- **IT Professionals and Developers**
 - Serves as a reference in developing HRIS solutions integrated with geofencing and biometric technologies.

REVIEW OF THE RELATED LITERATURE

- **Laudon & Laudon (2022)** – An integrated system acts as a "single source of truth," centralizing data across HR, service delivery, inventory, and procurement
- **Bangura (2024)** – HRIS has evolved from simple record-keeping into a centralized hub for all workforce-related data
- **Hemavathi & Chakravarthi (2024)** – Modern systems must follow "Privacy by Design," embedding RBAC and encryption from the start
- **ISO/IEC 25010:2023 (Rojas, 2025; Sarwosri, 2023)** – The gold-standard framework for software evaluation, covering 8 quality characteristics
- **RA 10173 (Data Privacy Act) & PD 442 (Labor Code)** – Legal mandates requiring secure data handling and accurate, timely wage payment

REVIEW OF THE RELATED STUDIES

- **Cay et al. (2022)** – Fingerprint attendance systems significantly reduced attendance fraud and "buddy punching"
- **German et al. (2021)** – A digitized tool-borrowing system (Quickip) eliminated bottlenecks, paper trails, and tracking errors
- **Ogundipe et al. (2024) & Shaikh et al. (2025)** – Automated payroll reduces processing time and boosts employee satisfaction/retention
- **Ariningsih (2025) & Jaquilmo et al. (2023)** – Digital ticketing systems improve accountability; "aging of transactions" is a key performance metric
- **Vilma & Booshnam (2025)** – Organizational performance only improves once employees fully accept the new system (change management)

METHODOLOGY

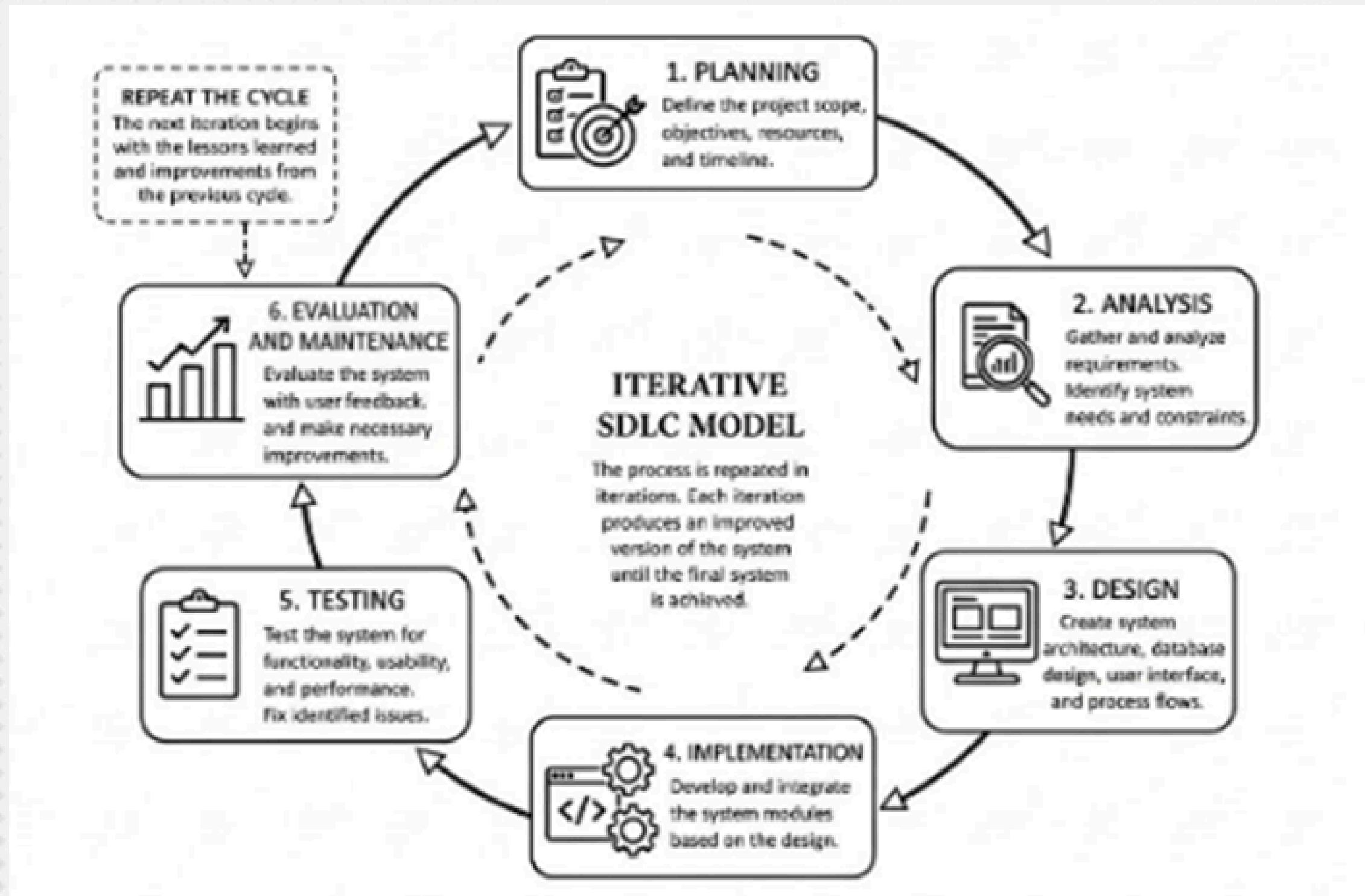
Research Design

Experimental quantitative research design Applied developmental research design

Iterative SDLC Model

Iterative SDLC model selection Planning and requirements analysis phases
Architecture design and system implementation Comprehensive unit and
field testing Pilot evaluation and ongoing maintenance

METHODOLOGY



System Core Features